EXPERIMENT 11

Routing at Network Layer.

Aim:

To simulate static Routing and RIP using cisco packet Tracer.

A) Static Routing.

Static routing is a method where routers are manually added to the router routing table.

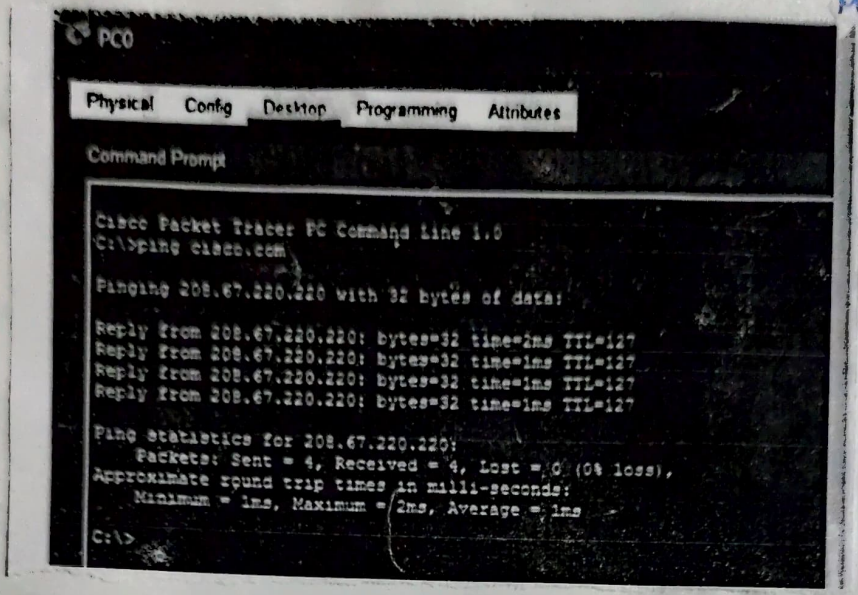
Procedure:

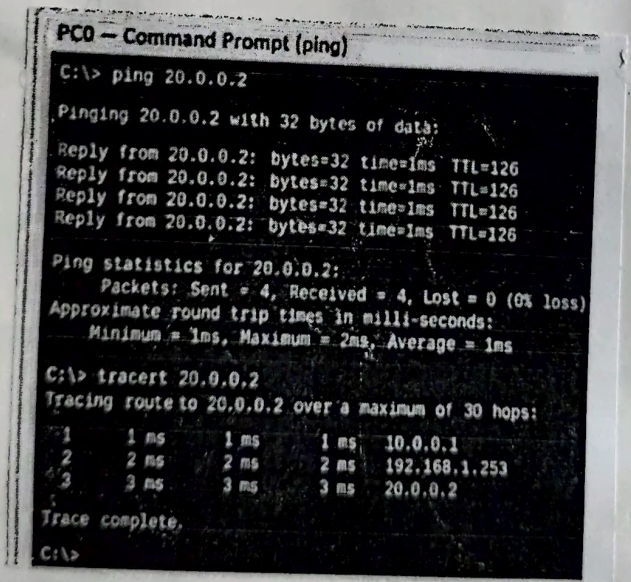
1) Open cisco packet Tracer.
2) Place required routers, switch and pc in workplace.
3) Connect the device using suitable cables.
4) assign Ip address to all pc, and router interface
5) Open the router CLI
6) Enter the foll command to configure static route.
7) Use foll command to check routing table. should IP route.
8) Use Ping command to check connectivity blw pcs.

B) RIP Routing.

RIP is a dynamic routing protocol. It allows routers to automatically exchange information about the network connected to them. RIP select routes based mainly on no of hops.

MIA


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping cisco.com
Pinging 208.67.220.220 with 32 bytes of data:
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Ping statistics for 208.67.220.220:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\>


PC0 — Command Prompt (ping)
C:\> ping 20.0.0.2
Pinging 20.0.0.2 with 32 bytes of data:
Reply from 20.0.0.2: bytes=32 time=1ms TTL=126
Reply from 20.0.0.2: bytes=32 time=1ms TTL=126
Reply from 20.0.0.2: bytes=32 time=1ms TTL=126
Reply from 20.0.0.2: bytes=32 time=1ms TTL=126
Ping statistics for 20.0.0.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\> tracert 20.0.0.2
Tracing route to 20.0.0.2 over a maximum of 30 hops:
1 1 ms 1 ms 1 ms 10.0.0.1
2 2 ms 2 ms 2 ms 192.168.1.253
3 3 ms 3 ms 3 ms 20.0.0.2
Trace complete.
C:\>

Procedure:

1) Create network with three routers and PCs shown in Packet tracer topology.

2) Assign Ip addresses to PCs and router interfaces

3) Configures the routes interface using

4) Enable RIP on router

5) Repeat RIP Configuration for other routers with their directly connected network.

7) Check the routing table using

show Ip address

8) Test Communicating using

Ping {destination Ip>

9) Send simple PDU from source PC to destination PC and observe Packet path.

Result:

Thus, static routing & RIP were successfully simulated using CISCO Packet Tracer.